

Pipelined processors

Example 1

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Computer Architectures

Example 2

Considering the following pseudo-code:

```
for (i = 0; i < 100; i++) {  
    Y[i] = X[i]2 + X[i] / Z[i]  
}
```

Suppose that vectors X[i] and Z[i] contain 100 FP numbers, were previously saved in memory, and Z[i] ≠ 0.

Assume the MIPS64 architecture presented below:

- Integer ALU: 1 clock cycle
- Data memory: 1 clock cycle
- FP multiplier unit: pipelined 4 stages
- FP arithmetic unit: pipelined 2 stages
- FP divider unit: not pipelined unit, requiring 4 clock cycles
- branch delay slot: 1 clock cycle (disabled)
- forwarding is enabled.

Example 2 – [cont]

- 1. Write an assembly program for the MIPS64 architecture able to perform the previously presented pseudo-code**
- 2. show the timing of the developed loop-based program and compute how many cycles does this program take to execute**
- 3. using all the static optimization techniques, re-write the developed code in order to eliminate the most data hazards**
- 4. show the timing development of the new optimized program and compute how many clock cycles does this program take to execute.**

1. Write an assembly program for the MIPS64 architecture able to perform the previously presented pseudo-code

```
.***** MIPS64          *****
;
;-----
;  $Y[i] = x[i]^2 + x[i] / z[i]$ 
;  $Z[i] \neq 0$ .
;-----

                .data
vetX:   .double 2, 4, 6, 8...
vetZ:   .double 1, 2, 3, 4...
vetY:   .double 0, 0, 0, 0...
```

1)

	.text	CC
MAIN:	daddui R1,R0,100	
	daddui R2,R0,vetX	
	daddui R3,R0,vetZ	
	daddui R4,R0,vetY	
loop:	l.d F1,0(R2)	
	l.d F2,0(R3)	
	mul.d F3,F1,F1	
	div.d F4,F1,F2	
	add.d F5,F3,F4	
	s.d F5,0(R4)	
	daddi R2,R2,8	
	daddi R3,R3,8	
	daddi R4,R4,8	
	daddi R1,R1,-1	
	bnez R1,loop	
	halt	

1)

	.text	CC
MAIN:	daddui R1,R0,100	5
	daddui R2,R0,vetX	1
	daddui R3,R0,vetZ	1
	daddui R4,R0,vetY	1
loop:	l.d F1,0(R2)	1
	l.d F2,0(R3)	1
	mul.d F3,F1,F1	4
	div.d F4,F1,F2	1
	add.d F5,F3,F4	2
	s.d F5,0(R4)	1
	daddi R2,R2,8	1
	daddi R3,R3,8	1
	daddi R4,R4,8	1
	daddi R1,R1,-1	1
	bnez R1,loop	2
	halt	1

2)

8

8

+

17 x 100

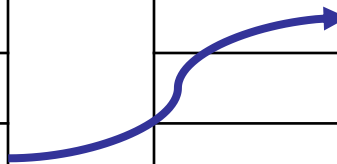
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3)

	.text	
MAIN:	daddui	R1,R0,100
	daddui	R2,R0,vetX
	daddui	R3,R0,vetZ
	daddui	R4,R0,vetY
loop:	l.d	F1,0(R2)
	l.d	F2,0(R3)
	mul.d	F3,F1,F1
	div.d	F4,F1,F2
	add.d	F5,F3,F4
	s.d	F5,0(R4)
	daddi	R2,R2,8
	daddi	R3,R3,8
	daddi	R4,R4,8

			cc
MAIN:	daddui	R1,R0,25	
	daddui	R2,R0,vetX	
	daddui	R3,R0,vetZ	
	daddui	R4,R0,vetY	
loop:	l.d	F1,0(R2)	
	l.d	F2,0(R3)	
	mul.d	F3,F1,F1	
	div.d	F4,F1,F2	
	daddi	R2,R2,8	
	add.d	F5,F3,F4	
	s.d	F5,0(R4)	
	daddi	R3,R3,8	
	daddi	R4,R4,8	



3)

[illegible]

3)

[illegible]

3+)

			CC	
MAIN:	daddui	R1,R0,25	5	8
	daddui	R2,R0,vetX	1	
	daddui	R3,R0,vetZ	1	
	daddui	R4,R0,vetY	1	
loop:	l.d	F1,0(R2)	1	
	l.d	F2,0(R3)	1	12
	mul.d	F3,F1,F1	4	
	div.d	F4,F1,F2	1	
	daddi	R2,R2,8	0	
	add.d	F5,F3,F4	2	
	s.d	F5,0(R4)	1	
	daddi	R3,R3,8	1	
	daddi	R4,R4,8	1	
	l.d	F1,0(R2)	1	12
	l.d	F2,0(R3)	1	
	mul.d	F3,F1,F1	4	
	div.d	F4,F1,F2	1	
	daddi	R2,R2,8	0	
	add.d	F5,F3,F4	2	
	s.d	F5,0(R4)	1	
	daddi	R3,R3,8	1	
	daddi	R4,R4,8	1	

	l.d	F1,0(R2)	1	12	4)
	l.d	F2,0(R3)	1		
	mul.d	F3,F1,F1	4		
	div.d	F4,F1,F2	1		
	daddi	R2,R2,8	0	8	+
	add.d	F5,F3,F4	2		
	s.d	F5,0(R4)	1		
	daddi	R3,R3,8	1		
	daddi	R4,R4,8	1	15	51 x 25
	l.d	F1,0(R2)	1		1283
	l.d	F2,0(R3)	1		
	mul.d	F3,F1,F1	4		
	div.d	F4,F1,F2	1		
	daddi	R2,R2,8	0		
	add.d	F5,F3,F4	2		
	s.d	F5,0(R4)	1		
	daddi	R3,R3,8	1		
	daddi	R1,R1,-1	1		
	daddi	R4,R4,8	1		
	bnez	R1,loop	1		
	halt		1		

4+) *branch delay slot* enabled

			cc
MAIN:	daddui	R1,R0,25	
	daddui	R2,R0,vetX	
	daddui	R3,R0,vetZ	
	daddui	R4,R0,vetY	
loop:	l.d	F1,0(R2)	
	l.d	F2,0(R3)	
	mul.d	F3,F1,F1	
	div.d	F4,F1,F2	
	add.d	F5,F3,F4	
	s.d	F5,0(R4)	

			cc
MAIN:	daddui	R1,R0,25	
	daddui	R2,R0,vetX	
	daddui	R3,R0,vetZ	
	daddui	R4,R0,vetY	
loop:	l.d	F1,0(R2)	
	l.d	F2,0(R3)	
	mul.d	F3,F1,F1	
	div.d	F4,F1,F2	
	add.d	F5,F3,F4	
	s.d	F5,0(R4)	
→	l.d	F1,8(R2)	
→	l.d	F2,8(R3)	
	mul.d	F3,F1,F1	
	div.d	F4,F1,F2	
	add.d	F5,F3,F4	
→	s.d	F5,8(R4)	

4+)

			CC	
MAIN:	daddui	R1,R0,25	5	8
	daddui	R2,R0,vetX	1	
	daddui	R3,R0,vetZ	1	
	daddui	R4,R0,vetY	1	
	daddui			
loop:	l.d	F1,0(R2)	1	10
	l.d	F2,0(R3)	1	
	mul.d	F3,F1,F1	4	
	div.d	F4,F1,F2	1	
	add.d	F5,F3,F4	2	
	s.d	F5,0(R4)	1	10
	l.d	F1,8(R2)	1	
	l.d	F2,8(R3)	1	
	mul.d	F3,F1,F1	4	
	div.d	F4,F1,F2	1	
	add.d	F5,F3,F4	2	
	s.d	F5,8(R4)	1	

	l.d	F1,16(R2)	1	10	4+)
	l.d	F2,16(R3)	1		8
	mul.d	F3,F1,F1	4		+
	div.d	F4,F1,F2	1		44 x 25
	add.d	F5,F3,F4	2		+
	s.d	F5,16(R4)	1	14	1
	l.d	F1,24(R2)	1		
	l.d	F2,24(R3)	1		1109
	mul.d	F3,F1,F1	4		
	div.d	F4,F1,F2	1		
	daddi	R2,R2,32	0	1	
	add.d	F5,F3,F4	2		
	s.d	F5,24(R4)	1		
	daddi	R1,R1,-1	1		
	daddi	R3,R3,32	1		
	bnez	R1,loop	1		
	daddi	R4,R4,32	1		
	Halt		1		